Ibn Tofail University

Analysis V

Problem Set III

Exercise 1:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$f(x,y) = \begin{cases} \dfrac{|xy|^\alpha}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}, \quad \alpha > 0.$$

Determine the values of α for which f is:

- continuous at $(0,0)$.
- differentiable at $(0,0)$.

Correction

Answer (no polar coordinates). The function f is

- continuous at $(0,0)$ exactly when $\alpha > 1$,
- differentiable at $(0,0)$ exactly when $\alpha > \frac{3}{2}$.

Proof. Using the elementary inequality $2|xy| \leq x^2 + y^2$ (which follows from $(|x| - |y|)^2 \geq 0$), we obtain for $(x,y) \neq (0,0)$

$$|xy|^\alpha \leq \left(\tfrac{x^2+y^2}{2}\right)^\alpha.$$

Thus

$$|f(x,y)| = \frac{|xy|^\alpha}{x^2+y^2} \leq 2^{-\alpha}(x^2+y^2)^{\alpha-1}.$$

Write $r^2 := x^2 + y^2$. The right-hand side equals $2^{-\alpha} r^{2\alpha-2}$.
Continuity. If $\alpha > 1$, then $2\alpha - 2 > 0$, so $r^{2\alpha-2} \to 0$ as $r \to 0$, and hence $f(x,y) \to 0$. Therefore f is continuous at $(0,0)$ for $\alpha > 1$.
To show this condition is sharp, restrict to the line $y = x$. For $x \neq 0$,

$$f(x,x) = \frac{|x \cdot x|^\alpha}{x^2+x^2} = \frac{|x|^{2\alpha}}{2x^2} = \tfrac{1}{2}|x|^{2\alpha-2}.$$

If $\alpha = 1$ this equals the constant $1/2$ (so the limit does not go to 0), and if $\alpha < 1$ the expression blows up as $x \to 0$. Thus continuity at the origin holds iff $\alpha > 1$.

Partial derivatives at the origin. For any h,

$$f(h,0) = f(0,h) = 0,$$

so

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(h,0)-0}{h} = 0, \qquad \frac{\partial f}{\partial y}(0,0) = 0.$$

Hence the only possible differential at the origin is the zero linear map $L \equiv 0$.
Differentiability. f is differentiable at the origin iff

$$\frac{f(x,y)-L(x,y)}{\sqrt{x^2+y^2}} = \frac{f(x,y)}{\sqrt{x^2+y^2}} \xrightarrow[(x,y)\to(0,0)]{} 0.$$

From the inequality above,

$$\left| \frac{f(x,y)}{\sqrt{x^2+y^2}} \right| \le 2^{-\alpha}(x^2+y^2)^{\alpha-\frac{3}{2}} = 2^{-\alpha} r^{2\alpha-3}.$$

Hence a sufficient condition for this quotient to tend to 0 is $2\alpha - 3 > 0$, i.e. $\alpha > \frac{3}{2}$.
To see necessity, again use the line $y = x$. For $x \neq 0$ we computed

$$f(x,x) = \tfrac{1}{2}|x|^{2\alpha-2}, \qquad \sqrt{x^2+x^2} = \sqrt{2}\,|x|,$$

so

$$\frac{f(x,x)}{\sqrt{x^2+x^2}} = \frac{\frac{1}{2}|x|^{2\alpha-2}}{\sqrt{2}\,|x|} = \frac{1}{2\sqrt{2}}\,|x|^{2\alpha-3}.$$

If $\alpha = \frac{3}{2}$ this ratio tends to the nonzero constant $\dfrac{1}{2\sqrt{2}}$; if $\alpha < \frac{3}{2}$ it blows up.
Therefore differentiability at $(0,0)$ fails unless $\alpha > \frac{3}{2}$.
Combining the sufficiency and necessity shows f is differentiable at $(0,0)$ exactly when $\alpha > \frac{3}{2}$.

Remark. For $(x,y) \neq (0,0)$ the formula defines a smooth function, so all issues are concentrated at the origin.

Exercise 2:

Study the existence and continuity of the first partial derivatives and the differentiability for the following functions:

$$f(x,y)=\begin{cases}\dfrac{x|y|}{\sqrt{x^2+y^2}} & \text{if } (x,y)\neq(0,0)\\ 0 & \text{if } (x,y)=(0,0)\end{cases};$$
$$f(x,y)=\begin{cases}\dfrac{xy}{|x|-y} & \text{if } y\neq|x|\\ 0 & \text{if } y=|x|\end{cases}$$
$$f(x,y)=\begin{cases}\dfrac{x^3-y^3}{x^2+y^2} & \text{if } (x,y)\neq(0,0)\\ 0 & \text{if } (x,y)=(0,0)\end{cases};$$
$$f(x,y)=\begin{cases}y^2\sin\left(\dfrac{x}{y}\right) & \text{if } y\neq 0\\ 0 & \text{if } y=0\end{cases}$$

Correction

(1) $f(x,y)=\begin{cases}\dfrac{x|y|}{\sqrt{x^2+y^2}} & \text{if } (x,y)\neq(0,0),\\ 0 & \text{if } (x,y)=(0,0).\end{cases}$

Continuity at $(0,0)$:

$$|f(x,y)|=\frac{|x||y|}{\sqrt{x^2+y^2}}\leq\frac{\frac{1}{2}(x^2+y^2)}{\sqrt{x^2+y^2}}=\frac{1}{2}\sqrt{x^2+y^2}\xrightarrow[(x,y)\to(0,0)]{}0.$$

Hence f is continuous at the origin.

Partial derivatives at $(0,0)$:

$$f(h,0)=0,\qquad f(0,h)=0,$$

so

$$\frac{\partial f}{\partial x}(0,0)=0,\qquad \frac{\partial f}{\partial y}(0,0)=0.$$

Continuity of partials: Away from the origin the function is smooth except for the absolute value. Compute for $y\neq 0$,

$$\frac{\partial f}{\partial x}=\frac{|y|}{\sqrt{x^2+y^2}}-\frac{x^2|y|}{(x^2+y^2)^{3/2}},\qquad \frac{\partial f}{\partial y}=\frac{x\,\mathrm{sign}(y)}{\sqrt{x^2+y^2}}-\frac{x|y|y}{(x^2+y^2)^{3/2}}.$$

The terms contain $\frac{|y|}{\sqrt{x^2+y^2}}$, which does not approach a single value as $(x,y)\to(0,0)$ along different paths. For instance, along $y=x$ one gets $|y|/\sqrt{x^2+y^2}=1/\sqrt{2}$, while along $y=0$ it is 0. Hence neither partial derivative is continuous at the origin.

Differentiability: For differentiability we need

$$\frac{|f(x,y)|}{\sqrt{x^2+y^2}}=\frac{|x||y|}{x^2+y^2}\xrightarrow[(x,y)\to(0,0)]{}0.$$

But along $y = x$, this equals $\frac{1}{2}$, a nonzero constant. Therefore f is *not differentiable* at the origin.

(2) $f(x,y) = \begin{cases} \dfrac{xy}{|x|-y}, & y \neq |x|, \\ 0, & y = |x|. \end{cases}$

Continuity: Along $y = 0$, $f(x,0) = 0$. Along $y = \frac{|x|}{2}$, $f(x,y) = \dfrac{x(\frac{|x|}{2})}{|x| - \frac{|x|}{2}} = \dfrac{|x|}{2}$. This tends to 0 as $x \to 0$. However, if we approach along $y = |x| - x^2$,

$$f(x, |x| - x^2) = \frac{x(|x| - x^2)}{|x| - (|x| - x^2)} = \frac{x(|x| - x^2)}{x^2} = |x|/x - |x|.$$

For $x > 0$ this equals $1 - x$, for $x < 0$ it equals $-1 - |x|$. The limit depends on the direction and thus does not exist. Therefore f is *not continuous at the origin.* Hence no differentiability is possible there.

(3) $f(x,y) = \begin{cases} \dfrac{x^3 - y^3}{x^2 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$

Continuity: We can factor:

$$f(x,y) = \frac{(x-y)(x^2 + xy + y^2)}{x^2 + y^2}.$$

Then

$$|f(x,y)| \leq |x-y| \frac{x^2 + |xy| + y^2}{x^2 + y^2} \leq 3|x-y|,$$

since $x^2 + |xy| + y^2 \leq 3(x^2 + y^2)/2$. Thus as $(x,y) \to (0,0)$, $|f(x,y)| \leq 3|x-y| \to 0$. Hence f is continuous at the origin.

Partial derivatives at $(0,0)$: For $h \neq 0$,

$$f(h,0) = \frac{h^3}{h^2} = h, \qquad f(0,h) = -h,$$

so

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(h,0)}{h} = 1, \qquad \frac{\partial f}{\partial y}(0,0) = \lim_{h \to 0} \frac{f(0,h)}{h} = -1.$$

Differentiability: Check whether

$$f(x,y) - x + y = o(\sqrt{x^2 + y^2}).$$

We compute

$$f(x,y) - (x-y) = \frac{x^3 - y^3 - (x-y)(x^2 + y^2)}{x^2 + y^2} = \frac{-xy(x-y)}{x^2 + y^2}.$$

Hence

$$|f(x,y)-(x-y)| \leq \frac{|x||y||x-y|}{x^2+y^2} \leq \frac{1}{2}|x-y|.$$

Thus

$$\frac{|f(x,y)-(x-y)|}{\sqrt{x^2+y^2}} \leq \frac{1}{2}\frac{|x-y|}{\sqrt{x^2+y^2}} \leq \frac{1}{\sqrt{2}}.$$

Since this bound tends to 0 as $(x,y) \to (0,0)$, the differential exists and equals

$$df_{(0,0)}(h,k) = h-k.$$

Hence f is differentiable at the origin.

(4) $f(x,y) = \begin{cases} y^2 \sin\left(\dfrac{x}{y}\right), & y \neq 0, \\ 0, & y = 0. \end{cases}$

Continuity:

$$|f(x,y)| \leq y^2 \quad \Rightarrow \quad f(x,y) \to 0 \text{ as } (x,y) \to (0,0).$$

So f is continuous.

Partial derivatives at the origin:

$$f(h,0) = 0 \Rightarrow f_x(0,0) = 0.$$

For f_y,

$$\frac{f(0,h)-f(0,0)}{h} = \frac{h^2\sin(0)}{h} = 0 \Rightarrow f_y(0,0) = 0.$$

Differentiability: We check

$$\frac{|f(x,y)|}{\sqrt{x^2+y^2}} \leq \frac{y^2}{\sqrt{x^2+y^2}} \leq |y| \xrightarrow[(x,y)\to(0,0)]{} 0,$$

so f is differentiable at the origin with differential $df_{(0,0)} = 0$.

Exercise 3:

Calculate $d_{(1,2)}f(x,y)$ and $d_{(2,3,4)}g(x,y,z)$ where $f(x,y) = \dfrac{x}{y^2}$ and $g(x,y,z) = \dfrac{z}{\sqrt{x^2+y^2}}$.

Correction

(a) Let $f(x,y) = \dfrac{x}{y^2}$. For $y \neq 0$, the partial derivatives are

$$\frac{\partial f}{\partial x}(x,y) = \frac{1}{y^2}, \qquad \frac{\partial f}{\partial y}(x,y) = -\frac{2x}{y^3}.$$

Hence the differential (total derivative) at a point (x, y) is

$$d_{(x,y)}f(h, k) = \frac{1}{y^2}\,h - \frac{2x}{y^3}\,k.$$

At the specific point $(1, 2)$:

$$\frac{\partial f}{\partial x}(1, 2) = \frac{1}{4}, \qquad \frac{\partial f}{\partial y}(1, 2) = -\frac{2}{8} = -\frac{1}{4},$$

so

$$\boxed{d_{(1,2)}f(h, k) = \frac{1}{4}h - \frac{1}{4}k\,.}$$

(b) Let $g(x, y, z) = \dfrac{z}{\sqrt{x^2 + y^2}}$. For $(x, y) \neq (0, 0)$:

$$\frac{\partial g}{\partial x} = -\frac{zx}{(x^2 + y^2)^{3/2}}, \quad \frac{\partial g}{\partial y} = -\frac{zy}{(x^2 + y^2)^{3/2}}, \quad \frac{\partial g}{\partial z} = \frac{1}{\sqrt{x^2 + y^2}}.$$

Thus

$$d_{(x,y,z)}g(h, k, \ell) = -\frac{zx}{(x^2 + y^2)^{3/2}}\,h - \frac{zy}{(x^2 + y^2)^{3/2}}\,k + \frac{1}{\sqrt{x^2 + y^2}}\,\ell.$$

At the point $(2, 3, 4)$, we have $x^2 + y^2 = 13$, so

$$\frac{\partial g}{\partial x} = -\frac{8}{13^{3/2}}, \quad \frac{\partial g}{\partial y} = -\frac{12}{13^{3/2}}, \quad \frac{\partial g}{\partial z} = \frac{1}{\sqrt{13}}.$$

Hence

$$\boxed{d_{(2,3,4)}g(h, k, \ell) = -\frac{8}{13^{3/2}}\,h - \frac{12}{13^{3/2}}\,k + \frac{1}{13^{1/2}}\,\ell\,.}$$

Jacobian-row form: $[\,-\frac{8}{13^{3/2}}\;,\; -\frac{12}{13^{3/2}}\;,\; \frac{1}{13^{1/2}}\,]$.

Exercise 4:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R}^2 \to \mathbb{R}^2$ be the functions defined by

$$f(x, y) = \sin(x^2 - y^2) \quad \text{and} \quad g(x, y) = (x + y, x - y).$$

1. Calculate the gradient of f at $(x, y) \in \mathbb{R}^2$.
2. Calculate the Jacobian matrix of g at $(x, y) \in \mathbb{R}^2$.
3. Deduce the Jacobian matrix of $f \circ g$ at $(x, y) \in \mathbb{R}^2$ and give the expression of $D(f \circ g)(x, y)(h, k)$.

Correction

(1) Gradient of f at $(x,y)\in\mathbb{R}^2$:
Compute the partial derivatives:

$$\frac{\partial f}{\partial x}(x,y)=\cos(x^2-y^2)\cdot 2x=2x\cos(x^2-y^2),$$

$$\frac{\partial f}{\partial y}(x,y)=\cos(x^2-y^2)\cdot(-2y)=-2y\cos(x^2-y^2).$$

Hence the gradient is

$$\boxed{\nabla f(x,y)=\big(2x\cos(x^2-y^2),\ -2y\cos(x^2-y^2)\big)}.$$

(2) Jacobian matrix of g at $(x,y)\in\mathbb{R}^2$:

$$g(x,y)=\begin{pmatrix}x+y\\x-y\end{pmatrix},\quad J_g(x,y)=\begin{pmatrix}\frac{\partial(x+y)}{\partial x} & \frac{\partial(x+y)}{\partial y}\\ \frac{\partial(x-y)}{\partial x} & \frac{\partial(x-y)}{\partial y}\end{pmatrix}=\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}.$$

(3) Jacobian matrix of $f\circ g$ and differential $D(f\circ g)(x,y)(h,k)$:
By the chain rule for functions $\mathbb{R}^2\to\mathbb{R}$ and $\mathbb{R}^2\to\mathbb{R}^2$:

$$D(f\circ g)(x,y)=\nabla f(g(x,y))\cdot J_g(x,y),$$

where $\nabla f(g(x,y))$ is a row vector and J_g is a 2×2 matrix.
Compute $\nabla f(g(x,y))$:

$$g(x,y)=(X,Y)=(x{+}y,x{-}y),\quad \nabla f(X,Y)=\big(2X\cos(X^2{-}Y^2),\ -2Y\cos(X^2{-}Y^2)\big)$$

Then

$$J_{f\circ g}(x,y)=\nabla f(g(x,y))\cdot J_g(x,y)=\big(2(X)\cos(X^2{-}Y^2),-2(Y)\cos(X^2{-}Y^2)\big)\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$$

Compute the product:

$$1^{st}\text{ component}=2X\cos(X^2-Y^2)+(-2Y)\cos(X^2-Y^2)=2(X-Y)\cos(X^2-Y^2)$$
$$2^{nd}\text{ component}=2X\cos(X^2-Y^2)+2Y\cos(X^2-Y^2)=2(X+Y)\cos(X^2-Y^2)$$

Substitute back $X=x+y$, $Y=x-y$:

$$X-Y=(x+y)-(x-y)=2y,\quad X+Y=(x+y)+(x-y)=2x.$$

Hence

$$\boxed{J_{f\circ g}(x,y)=\big(4y\cos(4xy),\ 4x\cos(4xy)\big)}.$$

Finally, for $(h,k)\in\mathbb{R}^2$:

$$\boxed{D(f\circ g)(x,y)(h,k)=4y\cos(4xy)\,h+4x\cos(4xy)\,k}.$$

Exercise 5:

Let $f : \mathbb{R}^n \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} \exp\left(\dfrac{1}{\|x\|^2 - 1}\right) & \text{if } \|x\| < 1 \\ 0 & \text{if } \|x\| \geq 1 \end{cases},$$

where $x = (x_1, \ldots, x_n)$ and $\|x\|^2 = \sum_{i=1}^{n} x_i^2$.

Show that f is of class C^1 on $\mathbb{R}^n$ and give the expression of its differential.

Correction

Step 1: Smoothness inside the ball $\|x\| < 1$.
Inside the unit ball, the function is a composition of smooth functions: $x \mapsto \|x\|^2$ and $t \mapsto 1/(t-1)$, then exponentiated. All derivatives exist and are smooth. Therefore $f \in C^\infty(\{x : \|x\| < 1\})$.

Step 2: Partial derivatives at $\|x\| = 1$.
For each $i = 1, \ldots, n$, the partial derivative for $\|x\| < 1$ is

$$\frac{\partial f}{\partial x_i}(x) = \exp\left(\frac{1}{\|x\|^2 - 1}\right) \cdot \frac{\partial}{\partial x_i}\left(\frac{1}{\|x\|^2 - 1}\right) = \exp\left(\frac{1}{\|x\|^2 - 1}\right) \cdot \frac{-2x_i}{(\|x\|^2 - 1)^2}.$$

Step 3: Continuity at $\|x\| = 1$.
As $\|x\| \to 1^-$, the term $\frac{-2x_i}{(\|x\|^2-1)^2}$ blows up, but the exponential term

$$\exp\left(\frac{1}{\|x\|^2 - 1}\right) \to 0 \quad \text{faster than any polynomial blow-up.}$$

Hence

$$\lim_{\|x\| \to 1^-} \frac{\partial f}{\partial x_i}(x) = 0.$$

For $\|x\| > 1$, $f \equiv 0$ and all partial derivatives are 0. Thus the partial derivatives are continuous across $\|x\| = 1$.

Step 4: Differential and class C^1.
The differential at any $x \in \mathbb{R}^n$ is

$$\boxed{df_x(h) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x)\, h_i = \begin{cases} \exp\left(\dfrac{1}{\|x\|^2 - 1}\right) \displaystyle\sum_{i=1}^{n} \frac{-2x_i}{(\|x\|^2 - 1)^2}\, h_i, & \|x\| < 1, \\ 0, & \|x\| \geq 1. \end{cases}}$$

Since the partial derivatives are continuous everywhere, $f \in C^1(\mathbb{R}^n)$.

Exercise 6:

Let f be the function defined by

$$f(x,y) = \begin{cases} xy\dfrac{x^2-y^2}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Calculate the partial derivatives at the points $(x,0)$ and $(0,y)$ with $x, y \in \mathbb{R}$. Calculate the second-order partial derivatives at $(0,0)$.

Correction

Step 1: Partial derivatives at $(x,0)$ and $(0,y)$.
(a) Along $y=0$: $f(x,0)=0$. Then

$$\frac{\partial f}{\partial x}(x,0) = \lim_{h\to 0} \frac{f(x+h,0)-f(x,0)}{h} = 0,$$

$$\frac{\partial f}{\partial y}(x,0) = \lim_{k\to 0} \frac{f(x,k)-f(x,0)}{k} = \lim_{k\to 0} \frac{xk(x^2-k^2)/(x^2+k^2)}{k} = \lim_{k\to 0} \frac{x(x^2-k^2)}{x^2+k^2} = x.$$

(b) Along $x=0$: $f(0,y)=0$. Then

$$\frac{\partial f}{\partial x}(0,y) = \lim_{h\to 0} \frac{f(h,y)-f(0,y)}{h} = \lim_{h\to 0} \frac{hy(h^2-y^2)/(h^2+y^2)}{h} = -y,$$

$$\frac{\partial f}{\partial y}(0,y) = \lim_{k\to 0} \frac{f(0,y+k)-f(0,y)}{k} = 0.$$

Step 2: Second-order partial derivatives at $(0,0)$.
(a) Mixed partial $\partial^2 f/\partial x\partial y$ at $(0,0)$:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = 0.$$

Then

$$\frac{\partial^2 f}{\partial y\partial x}(0,0) = \lim_{k\to 0} \frac{\frac{\partial f}{\partial x}(0,k) - \frac{\partial f}{\partial x}(0,0)}{k} = \lim_{k\to 0} \frac{-k-0}{k} = -1.$$

(b) Mixed partial $\partial^2 f/\partial y\partial x$ at $(0,0)$:

$$\frac{\partial f}{\partial y}(0,0) = 0, \quad \frac{\partial^2 f}{\partial x\partial y}(0,0) = \lim_{h\to 0} \frac{\frac{\partial f}{\partial y}(h,0) - \frac{\partial f}{\partial y}(0,0)}{h} = \lim_{h\to 0} \frac{h-0}{h} = 1.$$

(c) Pure second-order partials at $(0,0)$:

$$\frac{\partial^2 f}{\partial x^2}(0,0) = \frac{\partial^2 f}{\partial y^2}(0,0) = 0.$$

Step 3: Summary of partial derivatives.

$$f_x(x,0) = 0, \quad f_y(x,0) = x,$$
$$f_x(0,y) = -y, \quad f_y(0,y) = 0,$$
$$f_{xx}(0,0) = 0, \quad f_{yy}(0,0) = 0, \quad f_{xy}(0,0) = 1, \quad f_{yx}(0,0) = -1.$$

Remark: The mixed partials at the origin are not equal, which shows that f is not C^2 at $(0,0)$.

Exercise 7:

Let f be the function defined by $f(x) = \|x\|$, $\forall x \in \mathbb{R}^n$. Study the differentiability of f on $\mathbb{R}^n$.

Correction

Step 1: Differentiability for $x \neq 0$.
For $x \neq 0$, the norm function can be written as

$$f(x) = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

The partial derivatives are

$$\frac{\partial f}{\partial x_i}(x) = \frac{x_i}{\|x\|}, \quad i = 1, \ldots, n.$$

Hence the differential at $x \neq 0$ is

$$\boxed{df_x(h) = \sum_{i=1}^{n} \frac{x_i}{\|x\|} h_i = \frac{x \cdot h}{\|x\|}}.$$

This shows f is differentiable at every point $x \neq 0$.

Step 2: Differentiability at $x = 0$.
Assume f is differentiable at 0. Then there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}$ such that

$$\lim_{h \to 0} \frac{\|h\| - L(h)}{\|h\|} = 0.$$

But $\|h\|$ is positively homogeneous: along h with $\|h\| = 1$,

$$\frac{\|h\| - L(h)}{\|h\|} = 1 - L(h),$$

which must tend to 0 as $h \to 0$. This is impossible because L is linear, so $L(h)$ cannot equal 1 in all directions.

Conclusion:

$$f(x) = \|x\| \text{ is differentiable on } \mathbb{R}^n \setminus \{0\}, \text{ but not differentiable at } 0.$$